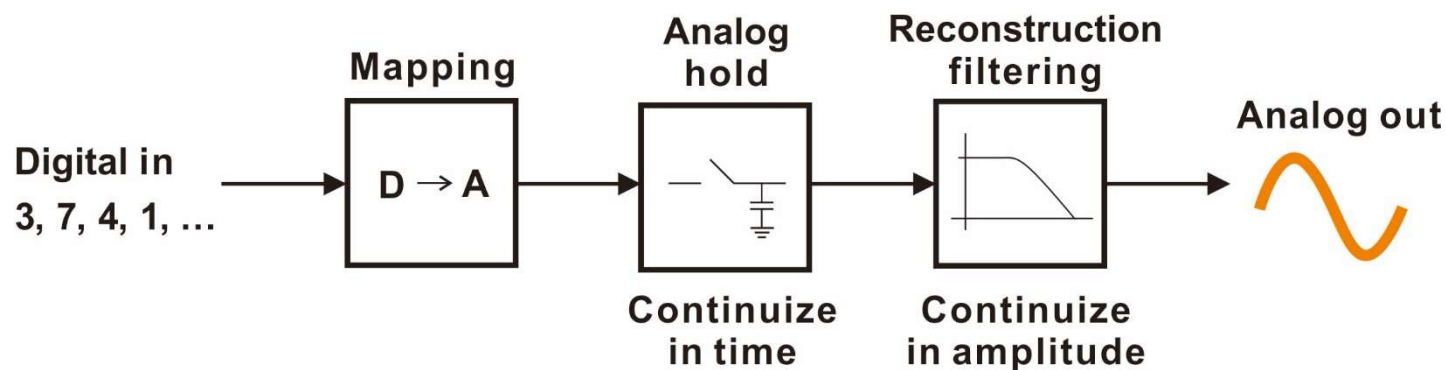


Oversampling DACs

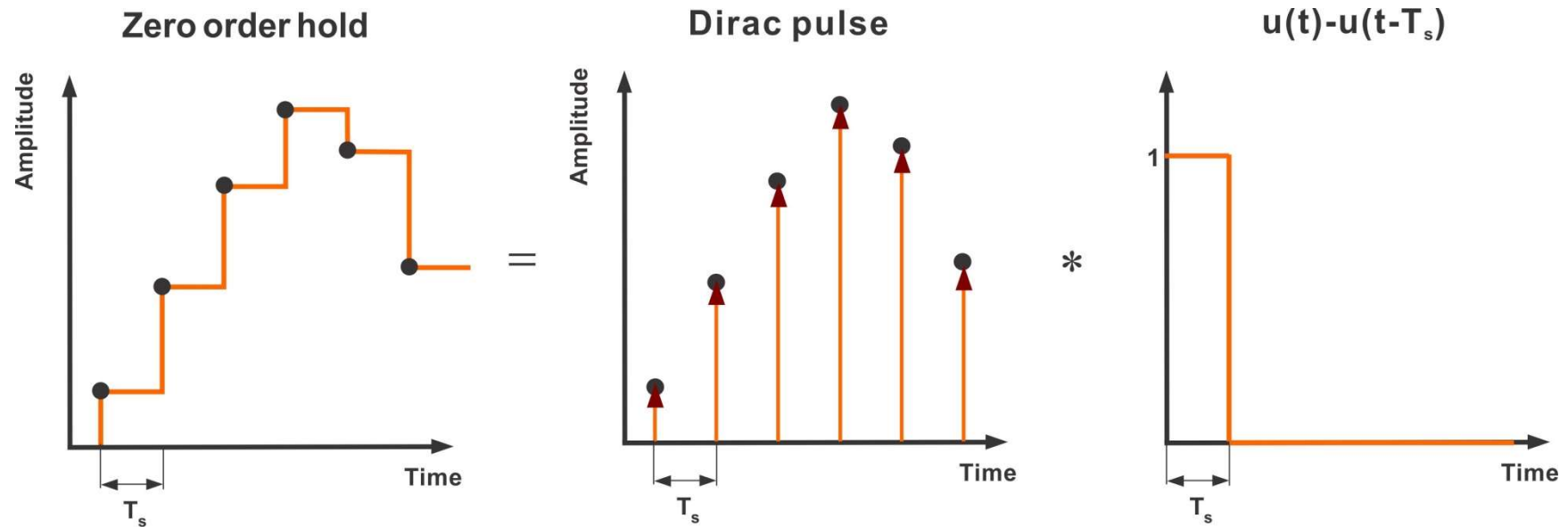
Nan Sun
Tsinghua University
nansun@tsinghua.edu.cn

Digital-to-Analog Conversion

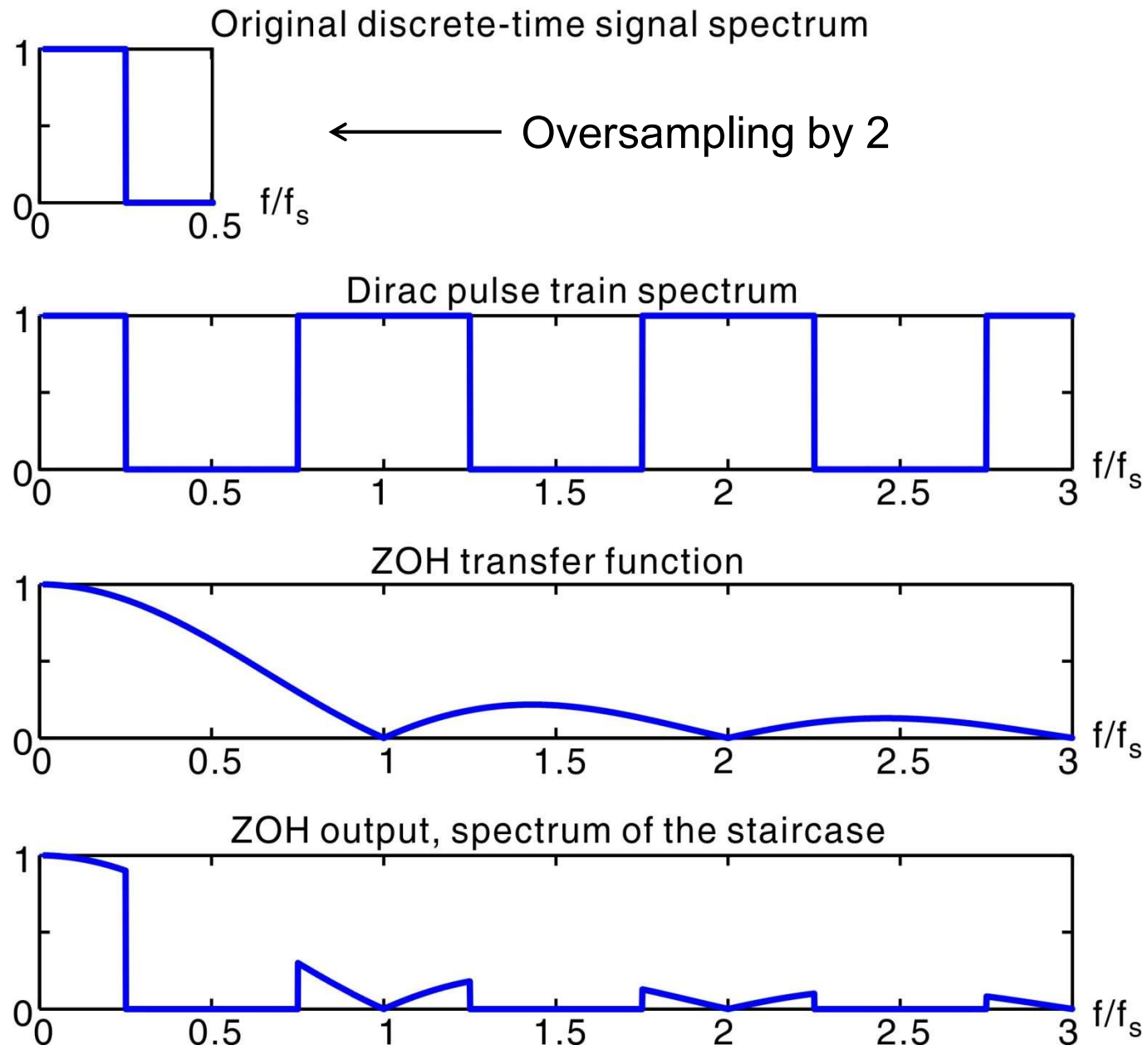


- What are the benefits for oversampling ($f_s = 2 \times \text{OSR} \times f_{\text{out}}$)?
 - Relax the requirement for the reconstruction filter
 - Use noise shaping to achieve 24-bit resolution with a 1-bit DAC
 - 1-bit DAC is simple and guaranteed linear!

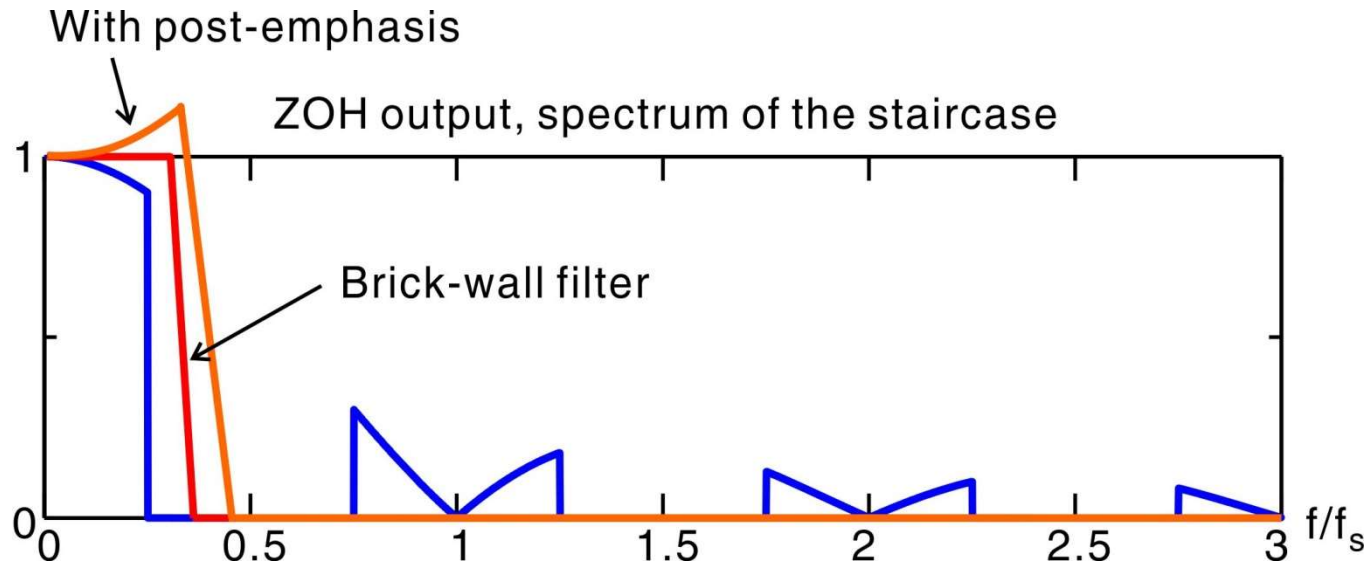
Zero-Order-Hold



$$X_{\text{ZOH}}(f) = \sum_{n=-\infty}^{\infty} X(f - nf_s) \times \frac{\sin(\pi f/f_s)}{\pi f/f_s} e^{-j\pi f/f_s}$$



Reconstruction Filter

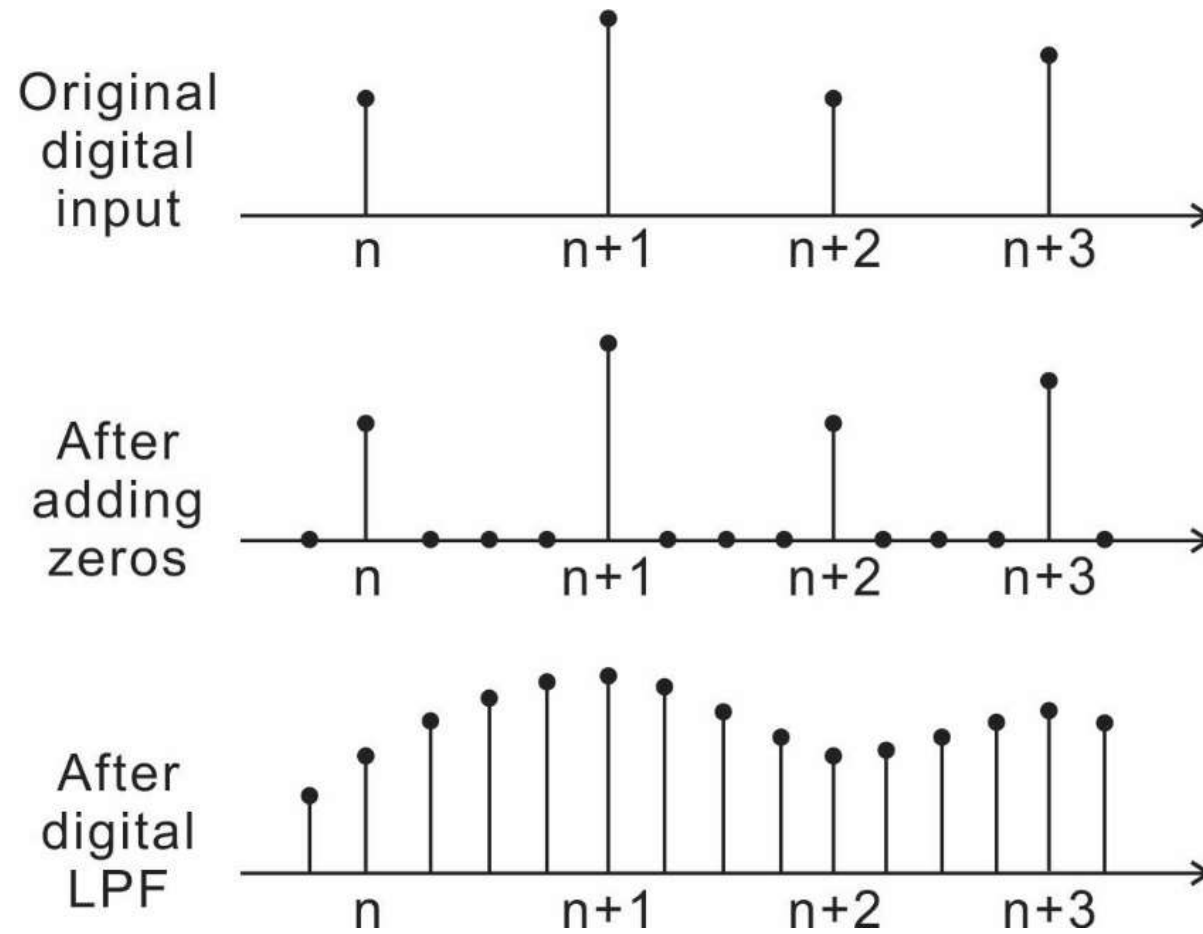


- We usually need a least (2x or 4x) oversampling even for Nyquist-rate DAC
 - Otherwise, the requirement for the reconstruction filter is too tough
 - Oversampling significantly relaxes the filter requirement

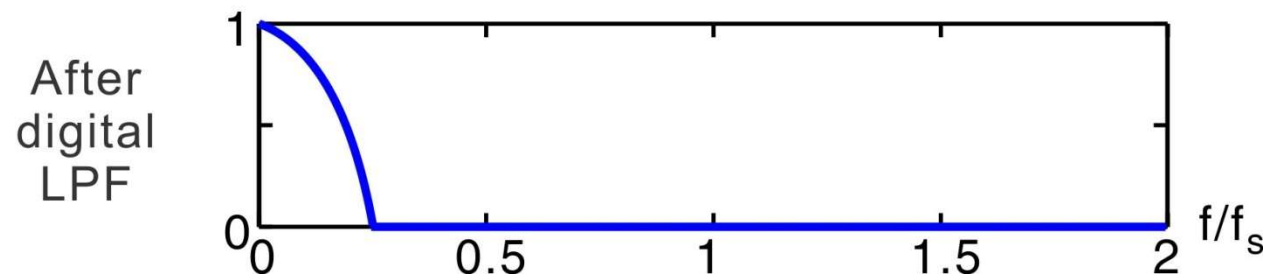
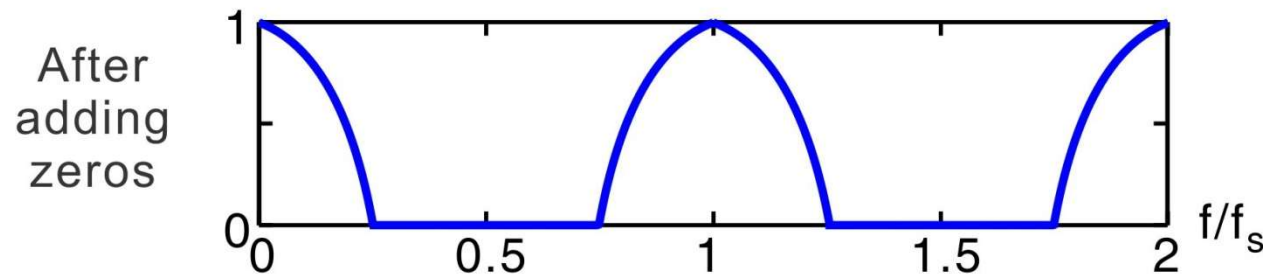
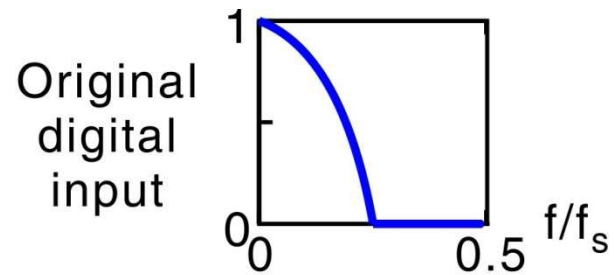
How to Oversample?

- How to oversample when the DAC digital input is at the Nyquist rate?
 - For an ADC whose input is continuous time, oversampling is easy as we can acquire the input at any given time instance
 - For a DAC, we only know the digital inputs at fixed time instances
- Several options
 - Perform an equivalent digital zero-order hold
 - Does this change the signal spectrum?
 - Linear interpolation between adjacent points
 - Does this distort the signal spectrum?
 - Adding zeros in between adjacent points and then digital LP filtering
 - Equivalent to sine wave reconstruction and then re-sampling

Interpolation



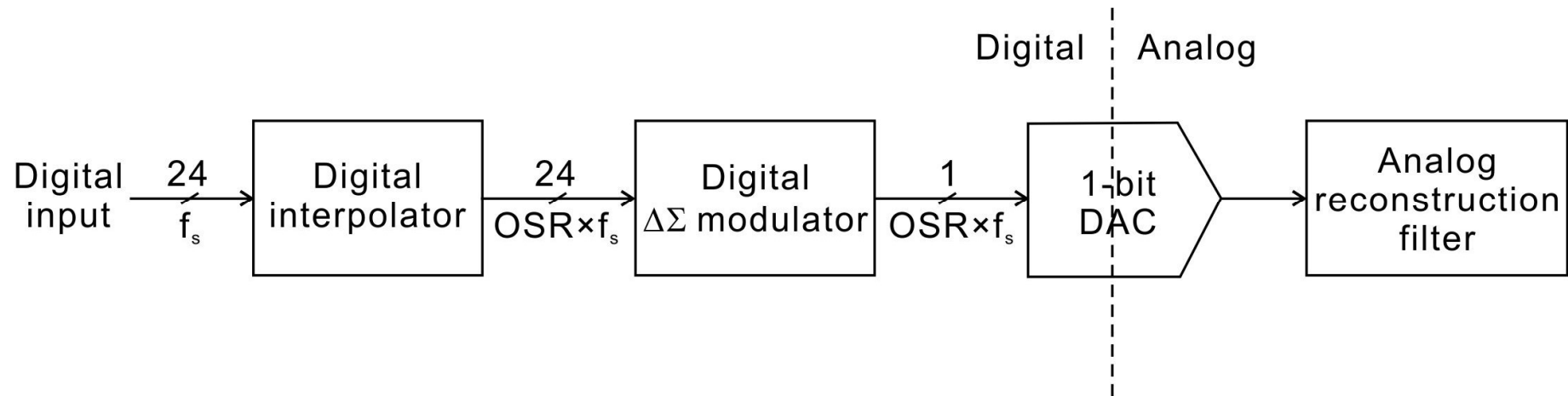
Frequency-Domain View



Benefits of Digital Oversampling

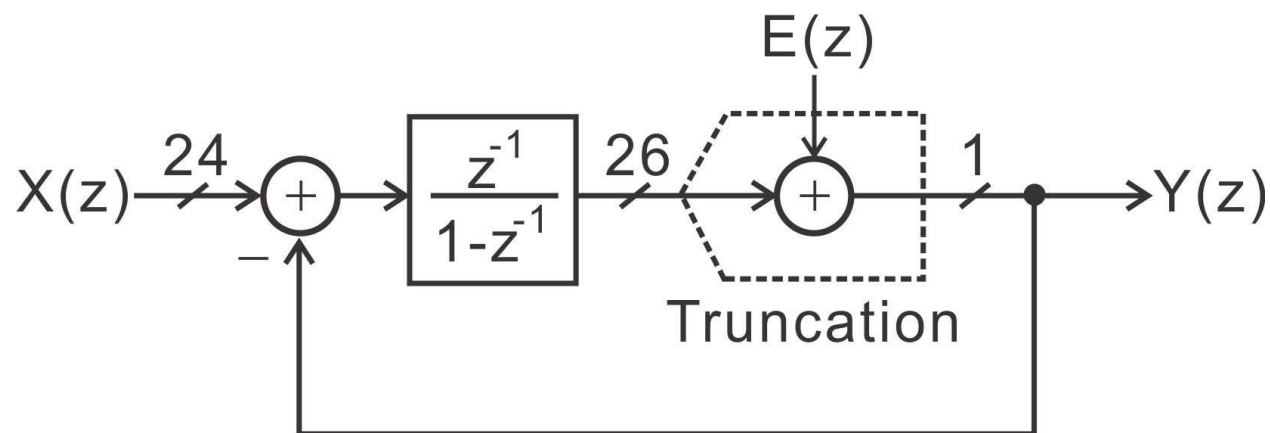
- The design of the analog reconstruction filter can be much simpler
 - The first few images are removed by digital low pass filtering
 - Analog re-construction is equivalent to oversampling by infinity
- Oversampling allows quantization noise shaping!
- Given 24-bit digital data, we want to convert it to an analog signal:
 - Option #1: design a 24-bit DAC
 - Complicated, area/power consuming, cannot guarantee linearity
 - Option #2: design a 16-bit DAC, and only use the first 16-bit data
 - Less complicated, less area/power, but less accuracy, nonlinear
 - Option #3, design a 1-bit DAC, and use noise shaping to move the truncation (quantization) error out of signal band
 - Easy to design, area/power efficient, and guaranteed linearity!

Oversampled DAC with Noise Shaping



- The digital $\Delta\Sigma$ modulator shapes the “truncation noise”
 - The truncation noise is equivalent to quantization noise by nature
 - Converting a fine-grain signal to a coarse-grain signal

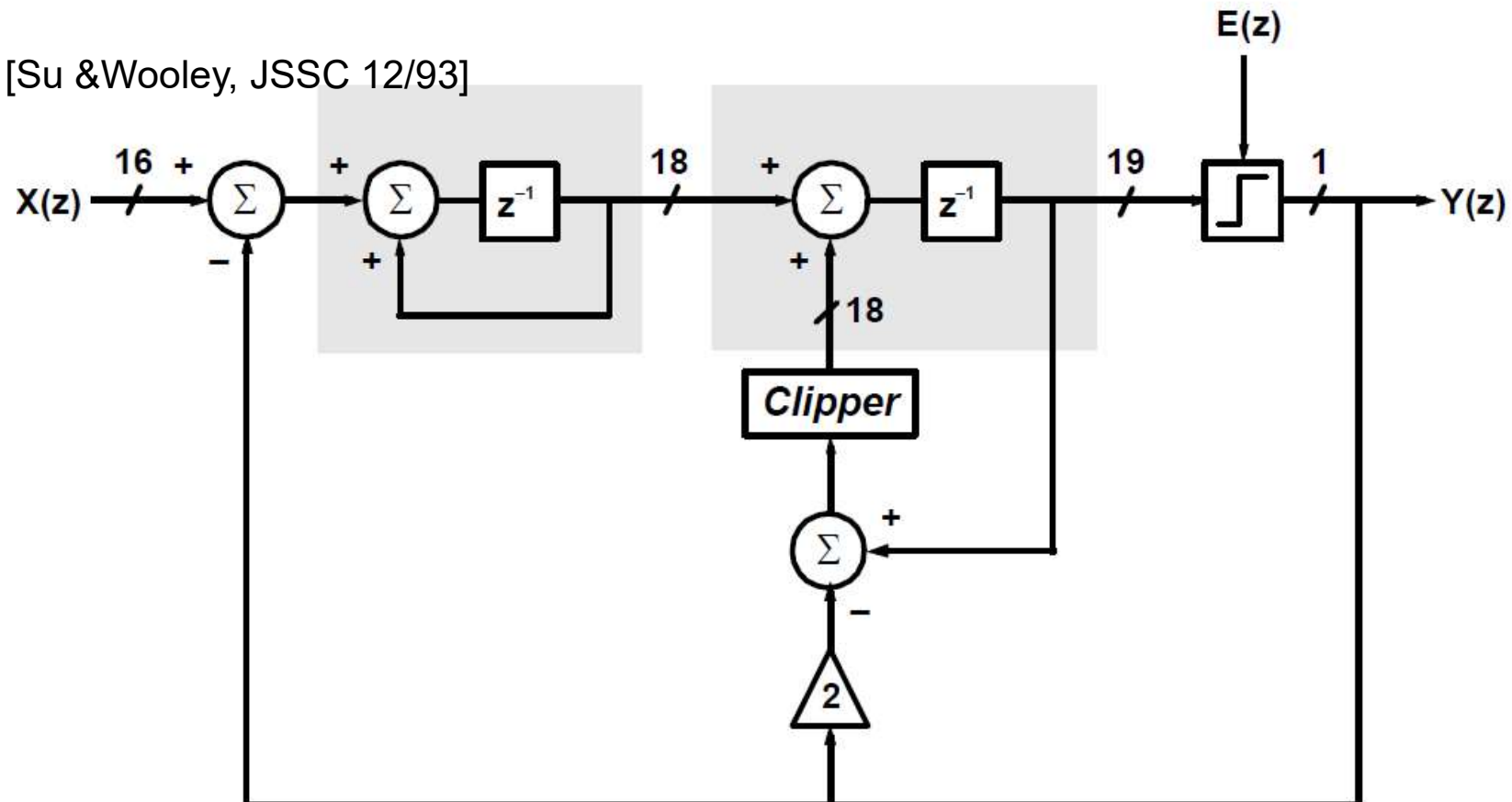
1st-Order Digital $\Delta\Sigma$ Modulator



- Same block diagram as an analog $\Delta\Sigma$ modulator
 - Intermediate nodes need to have more bits to ensure no overflow

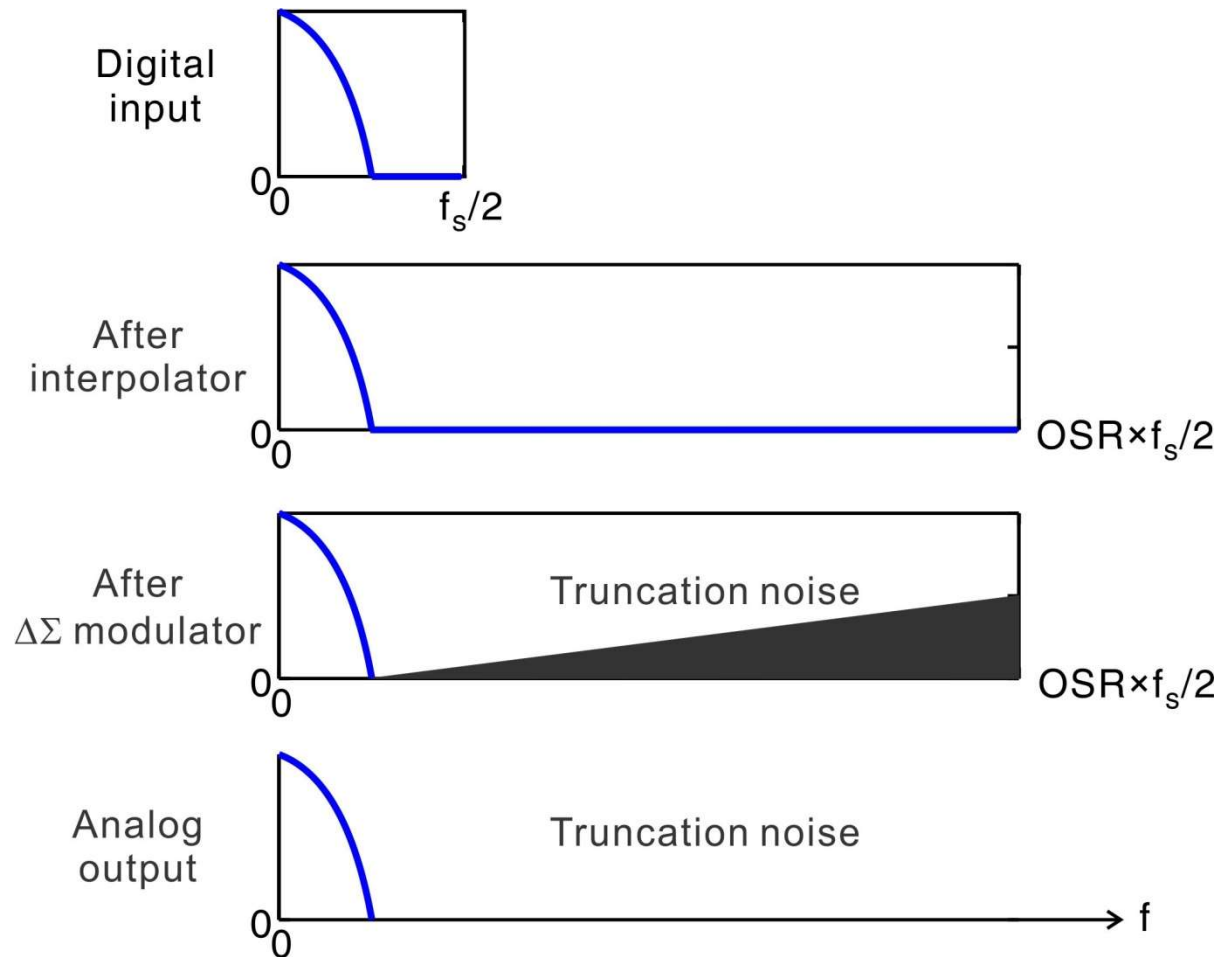
2nd-Order Digital $\Delta\Sigma$ Modulator

[Su & Wooley, JSSC 12/93]



$$Y(z) = z^{-2} X(z) + (1 - z^{-1})^2 E(z)$$

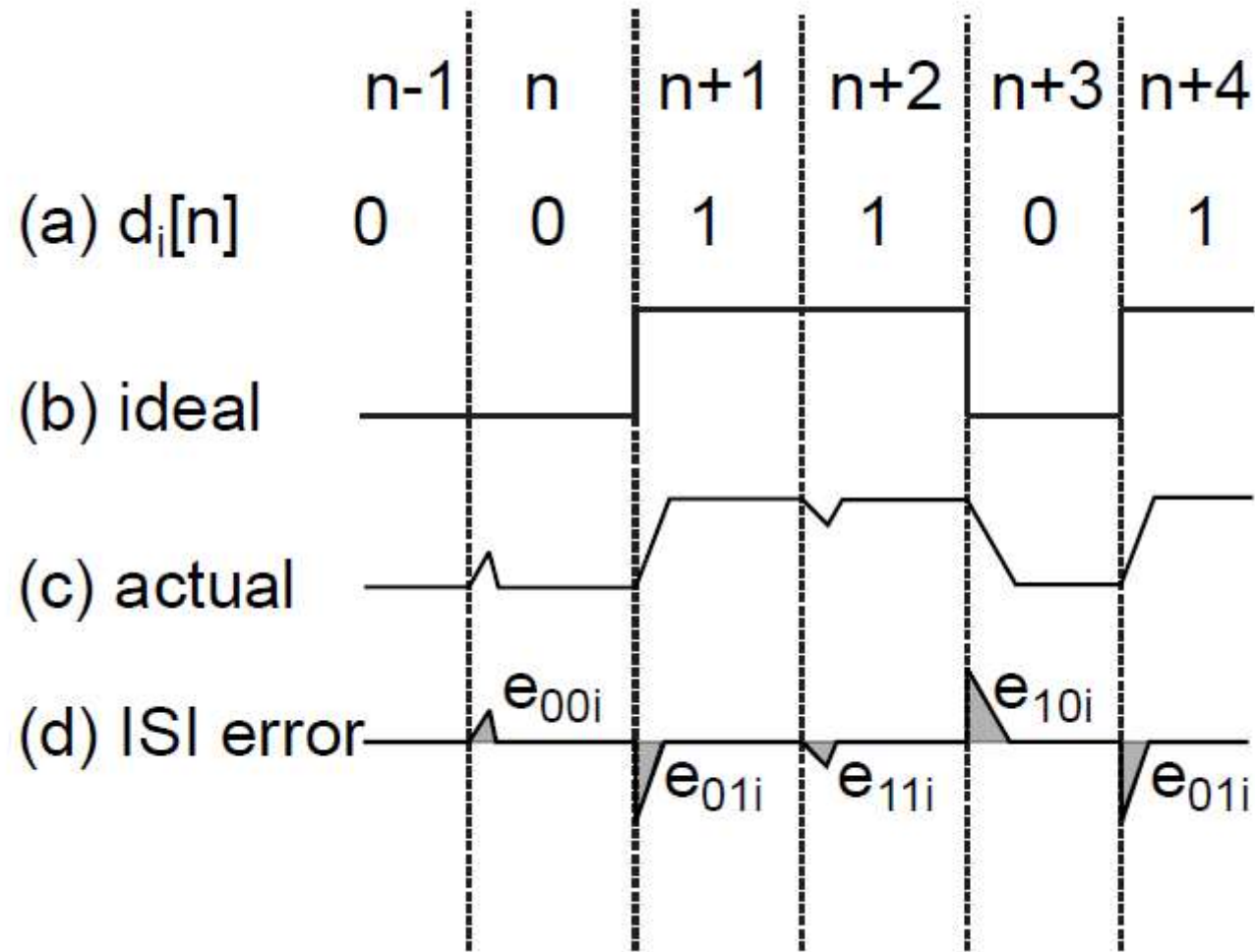
Spectra



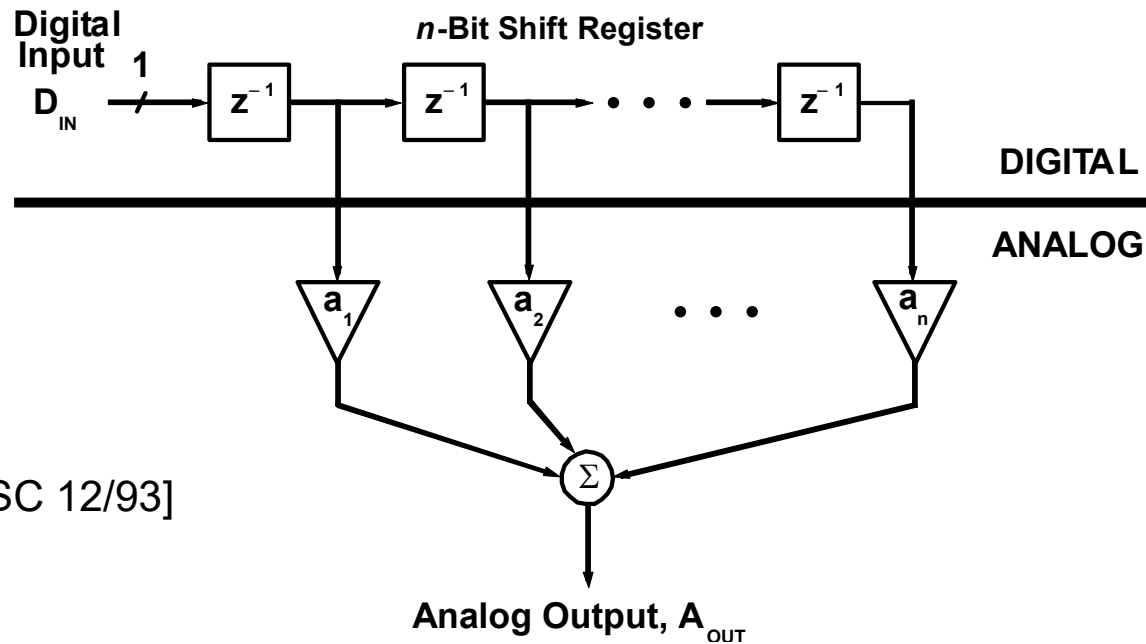
Problems with 1-bit DAC

- The DAC output has large jumps
 - It is sensitive to clock jitter
 - It is hard to guarantee linearity due to inter symbol interference, slewing, etc
 - 1-bit DAC is guaranteed linear only when we do not consider dynamic errors
- It would be nice if we can reduce the step size and apply some “digital” filtering before the DAC output goes to the analog reconstruction filter

ISI Error



Semi-Digital FIR Filter

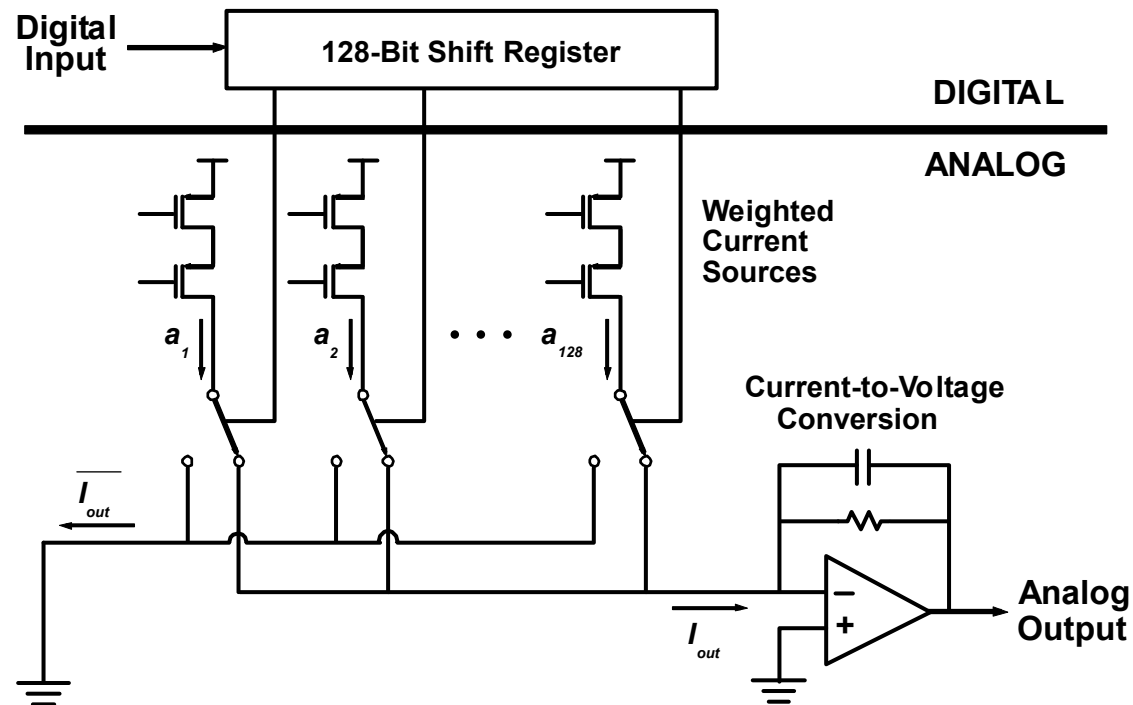


[Su & Wooley, JSSC 12/93]

$$A_{OUT}(z) = \underbrace{\left[a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + \dots + a_n z^{-n} \right]}_{H(z)} D_{IN}(z)$$

- An FIR filter is implemented by DAC weightings $\{a_i\}$

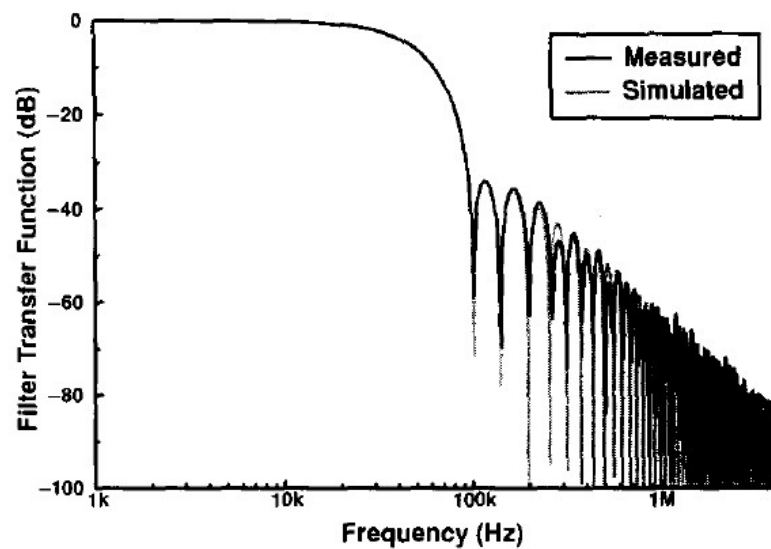
Circuit Implementation



- It seems that we still end up with 128 elements. Why is it better?
 - Variations in $\{a_i\}$ do not cause nonlinearity
 - They only change filter coefficients

Frequency Response

Reconstruction Filter



Output

